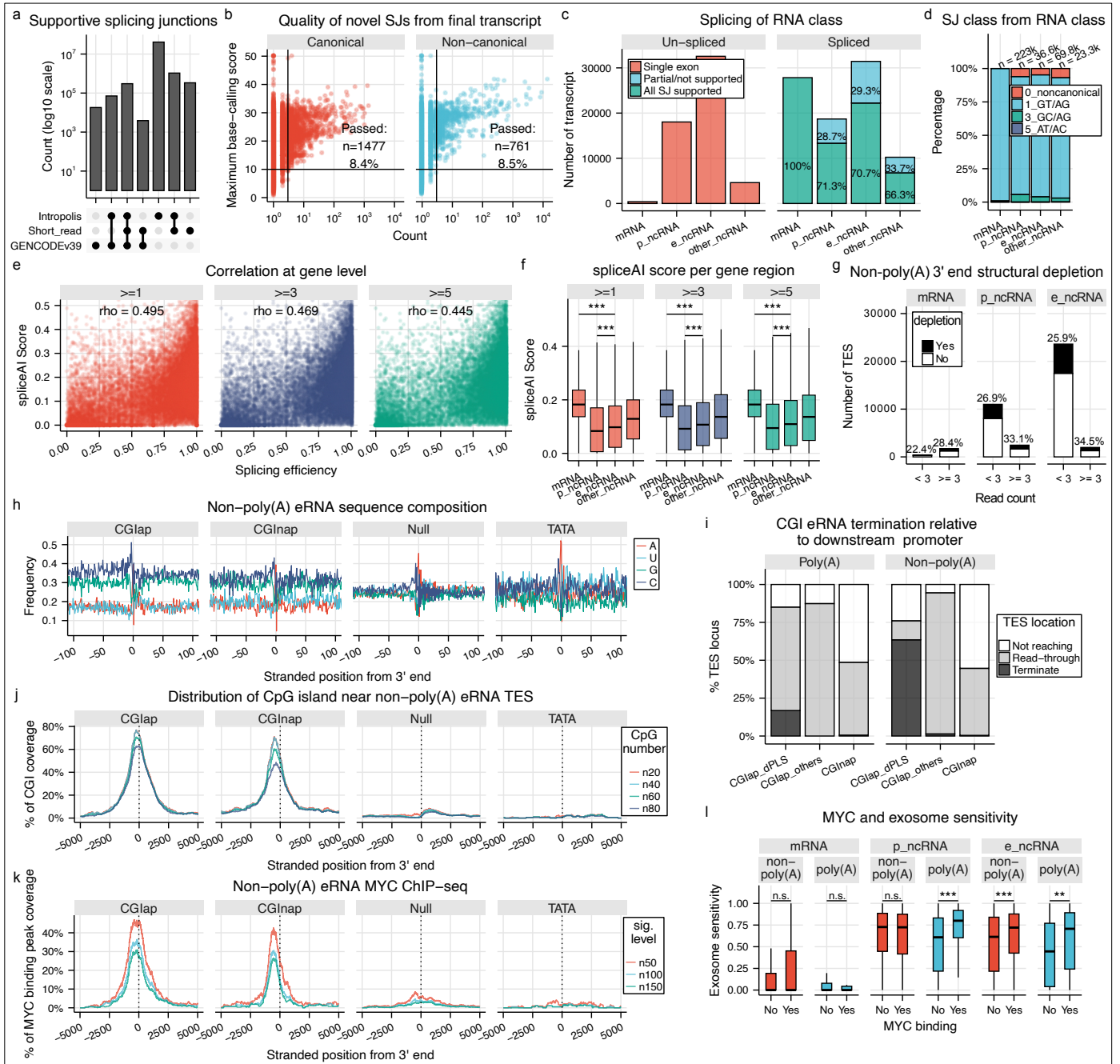




Extended Data Figure 7 | Splicing efficiency and eRNA transcription termination.

a, Number and percentage of splice junction (SJ) classes of all the SJs from external reference databases. “Short read” refers to confident SJs derived from short-read RNA-seq in the Neuron series. **b**, Novel SJs absent from the databases in (a) were classified as canonical or non-canonical. Plots show the relationship between the mean maximum base-calling score across the 6 nt surrounding each SJ and the SJ’s occurrence. SJs were considered technically confident if all 6 nt scored ≥ 10 and occurred ≥ 3 times. **c**, SJs from the final transcriptome were assigned to the four RNA classes, allowing for overlaps. For each class, the percentage of transcripts with and without fully supported SJs is shown. Positive SJ support was defined as either present in the databases in (a) or technically confident per (b). **d**, Total number of transcript models containing ≥ 1 SJ and percentage of each SJ class based on motif composition. **e**, Gene-level splicing efficiency compared to SpliceAI score across different occurrence thresholds. **f**, SpliceAI scores were summarized at the gene level and normalized by intron count. Gene models without SJs were excluded. SJs were filtered using three occurrence thresholds (sum of span and splice counts at both donor and acceptor sites). **g**, Evidence of RNA structural depletion (fold change $> 15\%$) at non-polyadenylated TESs, including both recursive and non-recursive TESs, based on RNAfold-derived hairpin scores. **h**, Nucleotide composition surrounding recursive transcription end sites (TESs; ≥ 3 read support) of non-poly(A) eRNAs, stratified by regulatory elements. **i**, Position of CGI enhancer-derived RNA TESs relative to the downstream promoter (SCREEN promoter cCRE). Not Reaching: TES is located > 500 nt upstream the promoter region, Terminate: TES is located within 500nt upstream or inside the promoter, Read-through: TES is located downstream of the promoter. **j**, Distribution of CpG islands proximal to TESs of non-polyadenylated RNA types, restricted to recursive TESs. **k**, Distribution of MYC binding sites (from ChIP-seq dataset GSM1505809) near TESs of non-polyadenylated RNA types, restricted to recursive TESs. **l**, Exosome sensitivity of transcript models was weighted and grouped by TES and compared between TES with and without MYC binding, stratified as different classes. Unless specified, statistical tests were two-sided Wilcoxon; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.